

MODULE 1 OF 5 • PRINTABLE NOTES

Literature Discovery and Understanding

Discover relevant scholarly literature, explore peer-reviewed research and understand key information from academic papers using AI-assisted search, synthesis and verification.

Boolean Search

Scopus AI

Publish or Perish

AI Discovery Tools

🎯 LEARNING OBJECTIVES

- Build precise Boolean search strings that combine synonyms with OR and concepts with AND.
- Locate literature across core databases (Scopus, Web of Science, Google Scholar) and subject databases.
- Use AI-powered discovery tools (Scopus AI, Semantic Scholar, Connected Papers, Elicit) to go beyond keyword search.
- Rank papers by citation impact using Publish or Perish (h-index, citations per year).
- Apply citation snowballing to find papers a keyword search would miss.
- Evaluate any source, especially an AI-suggested one, using the CRAAP test.

✓ BEFORE YOU START

- Bring 2–3 of your own research keywords or topics to use throughout every exercise in this module.
- Confirm your Scopus access (via UTM Library / EZproxy) works before the session starts.
- Download and install Publish or Perish (free) from harzing.com before the session.
- Have a free Google account ready for Google Scholar searches.



KEY CONCEPTS

Glossary of terms

Terms you will meet throughout this module, defined in plain language.

Boolean operators

The operators AND, OR and NOT, used to combine or exclude search terms.

h-index

A metric measuring both productivity and citation impact of an author's publications.

Citations per year

Total citations divided by years since publication, which normalises impact across paper age.

Semantic search

Search based on meaning and context rather than exact keyword matches.

Citation snowballing

Following a paper's citation trail (forward and backward) to find related work.

CRAAP test

A five-point check (Currency, Relevance, Authority, Accuracy, Purpose) for evaluating a source.



Trainer's Tip, Dr. Shahizan: Treat Scopus AI's synthesis as a map of the literature, not the destination. Always open at least the top three cited papers yourself and check whether the methodology matches what the AI described, before trusting the summary in your own writing.



AI TOOLS FOR THIS MODULE

AI tools and databases used in Module 1

TOOL	CATEGORY	WHAT IT DOES	WHERE TO ACCESS
Scopus	Core database	Comprehensive abstract & citation database of peer-reviewed journals, books and conference proceedings.	scopus.com (via UTM EZproxy)
Web of Science	Core database	Clarivate citation index and discovery platform for scholarly literature, proceedings, books and datasets.	webofscience.com (via UTM EZproxy)
Google Scholar	Free database	Free academic search engine covering papers, theses, books and abstracts.	scholar.google.com
Scopus AI	AI synthesis	Answers a plain-language research question with a synthesis drawn from Scopus-indexed literature.	Via UTM Library Scopus subscription
Publish or Perish	Citation metrics	Free desktop app that retrieves citations and calculates h-index, citations/year and more.	harzing.com/resources/publish-or-perish
Zeta Alpha	AI discovery	Neural / semantic search platform for technical and scientific knowledge discovery.	search.zeta-alpha.com
Semantic Scholar	AI discovery	Free AI-powered search for discovering papers and following citation connections.	semanticscholar.org
Connected Papers	Citation graph	Builds a visual graph of papers related to one seed publication.	connectedpapers.com
Elicit	AI literature search	Finds papers, supports study screening, and extracts data into structured tables.	elicit.com



HOW TO USE

Step-by-step tool guides



STEP-BY-STEP GUIDE

Scopus AI

Scopus AI lets you ask a plain-language research question and get a synthesised answer drawn from Scopus-indexed, peer-reviewed literature, along with suggested references and follow-up questions.

01

Access

Log in to Scopus AI through your institution's Scopus subscription (via the UTM Library) or a personal Scopus account with AI features enabled.

02

Ask

Type one natural-language question about your topic using the universal pattern below. Avoid Boolean operators; ask as you would ask a colleague.

03

Verify

Open the papers Scopus AI cites and check the abstract and methodology before treating any claim as established.

* UNIVERSAL QUESTION PATTERN

What are the [RESEARCH OBJECTIVE] regarding [TOPIC] in [CONTEXT OR POPULATION] over [OPTIONAL TIME BOUNDARY]?

Example: What are the emerging applications of explainable artificial intelligence for medical image diagnosis over the last five years?



STEP-BY-STEP GUIDE

Publish or Perish

A free desktop application that retrieves citations from Google Scholar, Scopus, Crossref and other sources, then calculates bibliometric indicators such as total citations, h-index and citations per year.

01

Search

Paste your Boolean search string into Publish or Perish, or search directly by author, topic or journal.

02

Rank

Sort results by citations per year to surface the most influential and most recent papers first.

03

Export

Export the reference list to BibTeX, RIS or CSV for your reference manager.



HANDS-ON PRACTICE

All exercises, with model answers

Every exercise below matches the hands-on activity on the module's web page. Attempt each one yourself first, then compare against the model answer.

EXERCISE 1.1

~10 min

Boolean Search String Builder

Group your keywords into up to three concepts. Combine synonyms within a concept with **OR**, and connect each concept with **AND**, producing a search string ready to paste into Scopus, Web of Science or Google Scholar's advanced search.

- ☐ Write down 2–3 concepts that make up your research topic (e.g., population, intervention, outcome).
- ☐ List every synonym or alternative wording for each concept, separated by commas.
- ☐ Combine synonyms within one concept using OR, in parentheses.
- ☐ Connect each concept's OR-group to the next with AND.
- ☐ Adjust field tags for your target database before pasting (e.g., TITLE-ABS-KEY in Scopus, TS= in Web of Science).

✓ WORKED EXAMPLE

Sample input — Concept 1: "student engagement, learner engagement" · Concept 2: "online learning, e-learning" · Concept 3: "higher education".

Generated string: ("student engagement" OR "learner engagement") AND ("online learning" OR "e-learning") AND "higher education"

EXERCISE 1.2

~10 min

Rank papers by citation impact with Publish or Perish

Use Publish or Perish to retrieve citations and calculate bibliometric indicators, helping you spot the most influential papers and authors before you commit to reading them.

- ☐ Install Publish or Perish (free) from harzing.com/resources/publish-or-perish before the session.
- ☐ Paste the search string from Exercise 1.1 into Publish or Perish, or search by author, topic or journal.
- ☐ Sort results by citations per year to surface the most influential and most recent papers first.
- ☐ Export the reference list to BibTeX, RIS or CSV for your reference manager.

✓ MODEL ANSWER — SAMPLE OUTPUT

- Search: ("student engagement" OR "learner engagement") AND ("online learning" OR "e-learning") — Google Scholar source, years 2018–2024.
- Papers retrieved: 187 · Total citations: 4,102 · h-index: 29 · Citations per year (average): 68.4.
- Top-ranked paper by citations/year: a 2021 study with 210 citations across 3 years (70/year) — a strong, recent candidate to read first.
- A citation count is one signal of impact, not a substitute for reading the paper: a highly-cited 2013 paper may simply be older, not necessarily more relevant to a 2024 research question.

EXERCISE 1.3

~15 min

Try Scopus AI with your own topic

Scopus AI lets you ask a plain-language research question and get a synthesised answer drawn from Scopus-indexed, peer-reviewed literature, along with suggested references and follow-up questions.

- ☐ Log in to Scopus AI through the UTM Library subscription or a personal Scopus account with AI features enabled.
- ☐ Ask Scopus AI your main research question using the universal pattern (see the tool guide above), with your own topic.
- ☐ Note down 3 papers Scopus AI surfaces that look genuinely relevant.
- ☐ Ask one follow-up question to explore a gap, trend or contradiction.
- ☐ Open one suggested paper and confirm the abstract actually supports what Scopus AI said.

✓ MODEL ANSWER – SAMPLE INTERACTION

- Question asked: "What are the emerging applications of explainable artificial intelligence for medical image diagnosis over the last five years?"
- Synthesis returned: three emerging application areas (radiology triage, histopathology classification, retinal disease screening), each with 2–4 supporting citations.
- 3 relevant papers noted: Lim et al. 2022 (systematic review), a 2023 radiology paper, and a 2024 histopathology paper.
- Follow-up question: "What methodological limitations are most commonly reported across these studies?" – surfaced a recurring theme of small, single-institution datasets.
- Verification: opened the 2023 radiology paper directly and confirmed its abstract matched the claim Scopus AI attributed to it, before noting it down as reliable.

EXERCISE 1.4

~15 min

Citation snowballing

Snowballing finds papers a keyword search misses, by following the citation trail of one strong paper instead of guessing more search terms.

- ☐ Pick one strong seed paper from your search results, ideally one that is highly relevant and reasonably well cited.
- ☐ Snowball forward: on Google Scholar, open the seed paper and click "Cited by" to see newer papers that cite it. Note 2–3 that look relevant.
- ☐ Snowball backward: open the seed paper's own reference list and note 2–3 references that look relevant to your topic.
- ☐ Record each new paper in the table, then repeat one more round from your best new find if time allows.

ROUND	DIRECTION	PAPER (AUTHOR, YEAR)	WHY IT IS RELEVANT
1	Forward (cited by)		
1	Backward (reference list)		
2			

✓ MODEL ANSWER – WORKED EXAMPLE

- Seed paper: "Explainable AI for Medical Image Diagnosis: A Systematic Review" (Lim et al., 2022).
- Forward (cited by): a 2024 paper applying the same explainability framework to radiology datasets – relevant because it shows how the seed paper's method has since been extended.
- Backward (reference list): the earlier paper that first proposed the explainability technique the seed paper builds on – relevant because it is the foundational source the seed paper depends on.
- Common mistake to avoid: recording every citing or cited paper without checking relevance first. Snowballing should still be selective – skim the title and abstract before adding a paper to the table.

EXERCISE 1.5

~15 min

Compare database coverage

No single database indexes everything. Running the same search in more than one database shows you what you would have missed by relying on just one.

- ☐ Take the search string from Exercise 1.1 and run it, adjusting field tags as needed, in two or three databases (e.g., Scopus, Web of Science, Google Scholar, or a subject database from your field).
- ☐ Record the total result count and the top result in each database.
- ☐ Check whether the top results overlap across databases, or whether each surfaces different papers.
- ☐ Note one likely reason for any difference (e.g., subject coverage, indexing lag, grey literature included or excluded).

DATABASE	TOTAL RESULTS	TOP RESULT	NOTABLE DIFFERENCE
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✓ MODEL ANSWER — SAMPLE FILLED TABLE

Scopus — 214 results, top result Smith 2021, notably strong in engineering & computer science.

Google Scholar — 1,860 results, top result Lee 2022, includes theses & preprints Scopus does not index.

Web of Science — 178 results, top result Smith 2021 (overlaps with Scopus), narrower coverage of conference proceedings.

Likely reason for the difference: Google Scholar's much larger count reflects its inclusion of grey literature (theses, preprints, non-peer-reviewed reports), while Scopus and Web of Science apply stricter peer-review and source-list criteria, trading recall for precision.

EXERCISE 1.6

~10 min

Check a source with the CRAAP test

Before you rely on a source, especially one an AI tool suggested, run it through the CRAAP test: Currency, Relevance, Authority, Accuracy, Purpose.

- ☐ Currency: when was it published or last updated? Is that recent enough for your topic?
- ☐ Relevance: does it actually address your research question, or just share a keyword?
- ☐ Authority: who are the authors, and what is their expertise or affiliation? Is the venue peer-reviewed?
- ☐ Accuracy: is it well-referenced, and can you verify its claims or data against other sources?
- ☐ Purpose: why was it written? Is there a bias, funding source or agenda you should account for?

✓ MODEL ANSWER — WORKED EXAMPLE

- Source: a 2024 conference paper on AI-assisted grading, surfaced by Scopus AI.
- Currency — published 2024, well within a fast-moving field: pass.
- Relevance — directly addresses grading accuracy, matching the research question closely: pass.
- Authority — authors affiliated with a recognised university CS department, published in an IEEE conference: pass.
- Accuracy — 34 references, methodology and dataset described in enough detail to check claims: pass.
- Purpose — no funding disclosure found; flagged as "verify further" rather than assumed neutral.
- Overall recommendation: cite with caution — verify the funding/purpose question before treating its conclusions as bias-free.

WHY USE AI HERE



Traditional vs AI-assisted approach

Task	Traditional Approach	AI-Assisted Approach
Building a search string	Trial-and-error keyword combinations, one database at a time	Search String Builder generates a complete Boolean string instantly
Finding related papers	Follow each reference list manually, one citation at a time	Connected Papers or Zeta Alpha map related work instantly as a graph
Judging a paper's impact	Look up citation counts one paper at a time across separate sites	Publish or Perish calculates h-index and citations per year in one pass
Understanding findings across papers	Read every abstract individually and synthesise by hand	Scopus AI synthesises an answer across dozens of papers at once

The trade-off: AI tools save time on breadth; you still verify depth. Always open the underlying papers before treating a synthesised claim as established.

COMMON PITFALLS



Things to verify before you trust the output

- ☐ Trusting a Scopus AI synthesis without opening at least the top 2–3 cited papers to check the methodology matches the claim.
- ☐ Treating a high citation count as proof of quality – an older paper accumulates citations simply by existing longer.
- ☐ Recording every citing/cited paper during snowballing without first checking relevance – this defeats the purpose of being selective.
- ☐ Relying on a single database and assuming its coverage is complete – always cross-check against at least one more source.
- ☐ Skipping the CRAAP test's Purpose question, especially for AI-surfaced sources where funding or agenda is not obvious at a glance.
- ☐ Forgetting to adjust Boolean field-tag syntax between databases (TITLE-ABS-KEY in Scopus vs TS= in Web of Science).

✦ Key Takeaways

- A good Boolean search string groups synonyms with OR and connects concepts with AND.
- Start with core multidisciplinary databases, then widen with AI-powered discovery tools.
- Citation metrics (Publish or Perish) rank impact, but never replace reading the paper.
- Always verify a Scopus AI synthesis against the original, cited publications.
- Citation snowballing and database-coverage comparisons catch what a single keyword search misses.

➤ **What's next:** Have a shortlisted topic and a handful of strong papers? Module 2 helps you turn that into a clear, FINER-checked research question.



FOR YOUR NOTES

Session Notes

A large rectangular area with horizontal lines, intended for taking session notes.



FOR YOUR NOTES

Session Notes (continued)